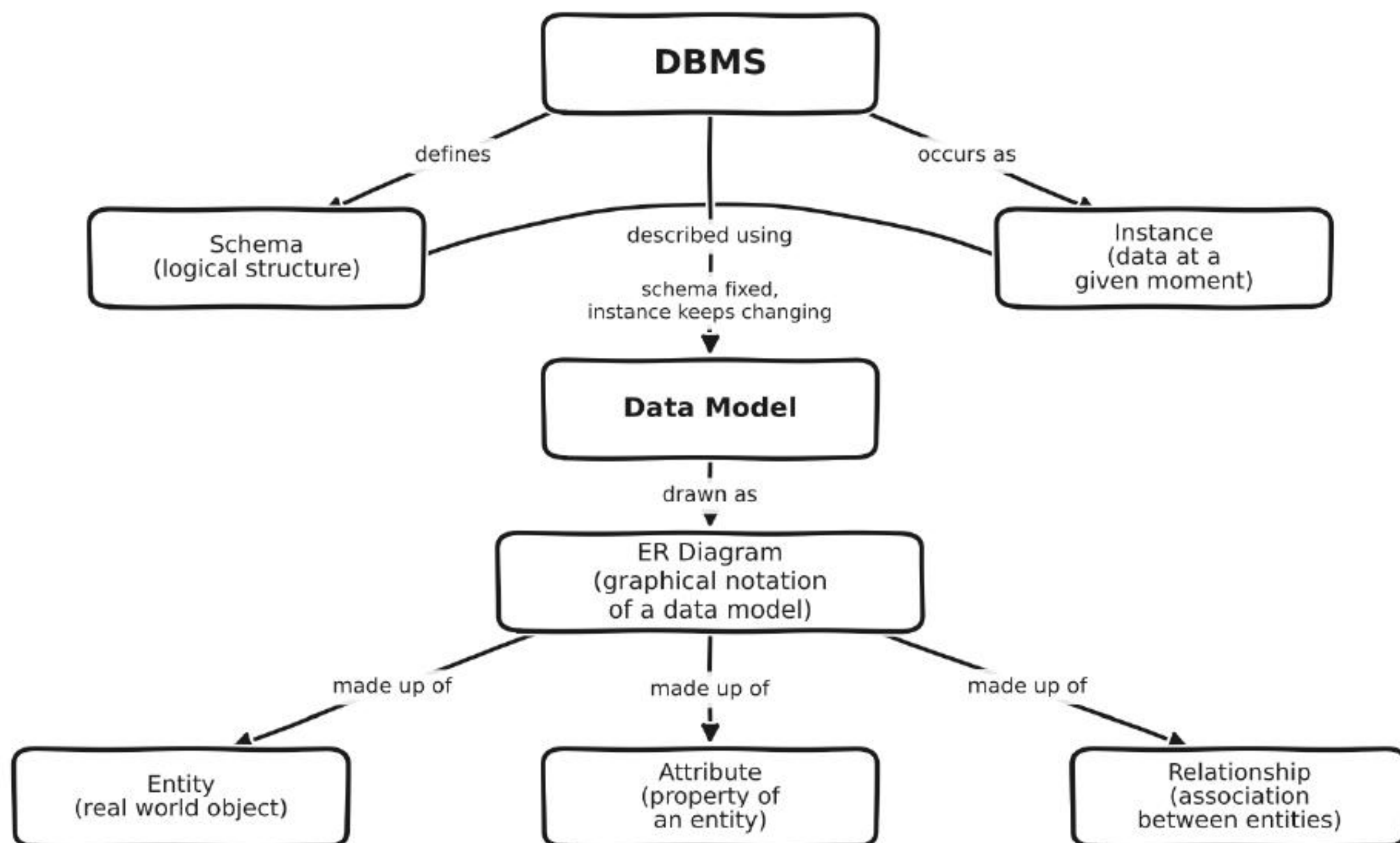
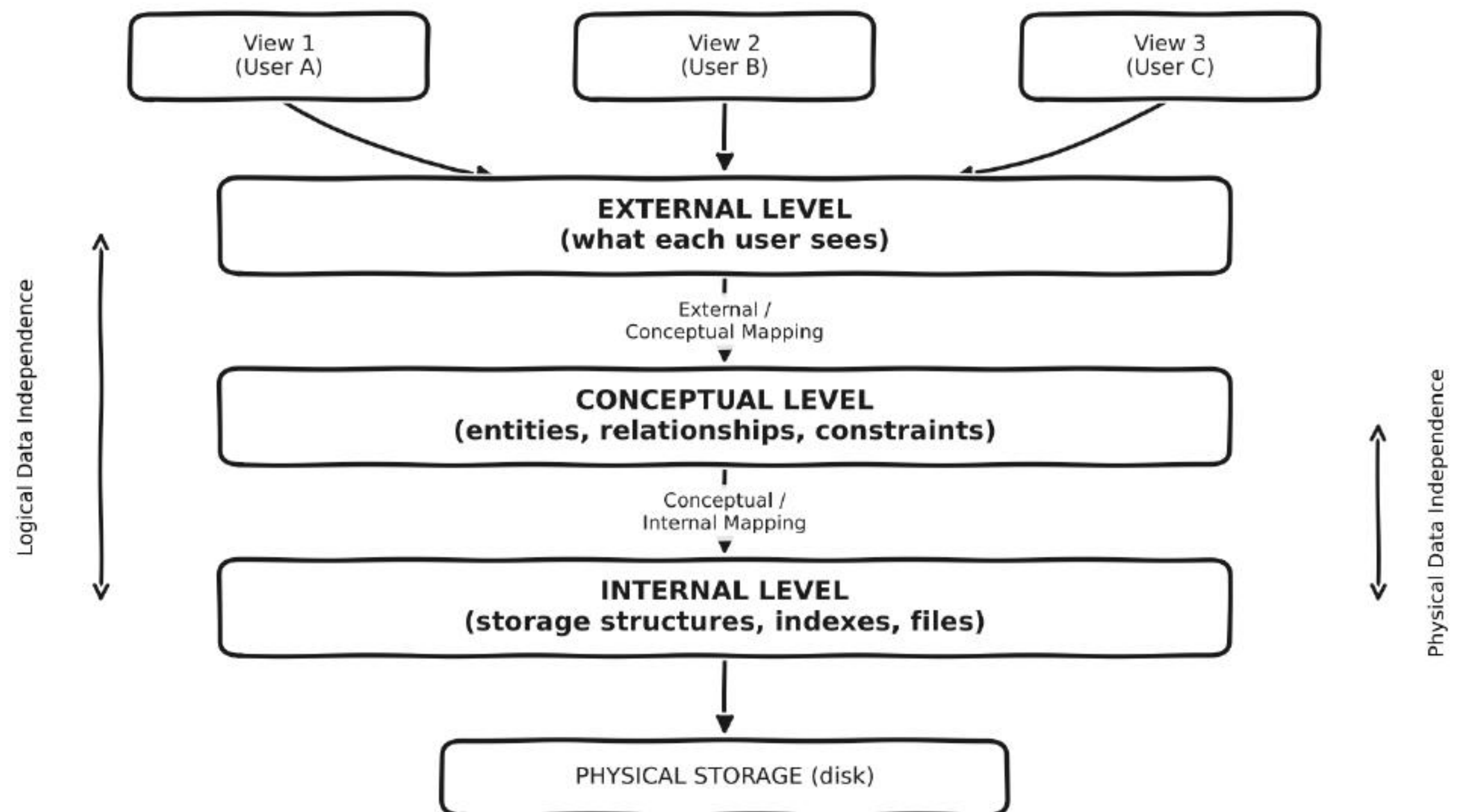


Q1. Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.

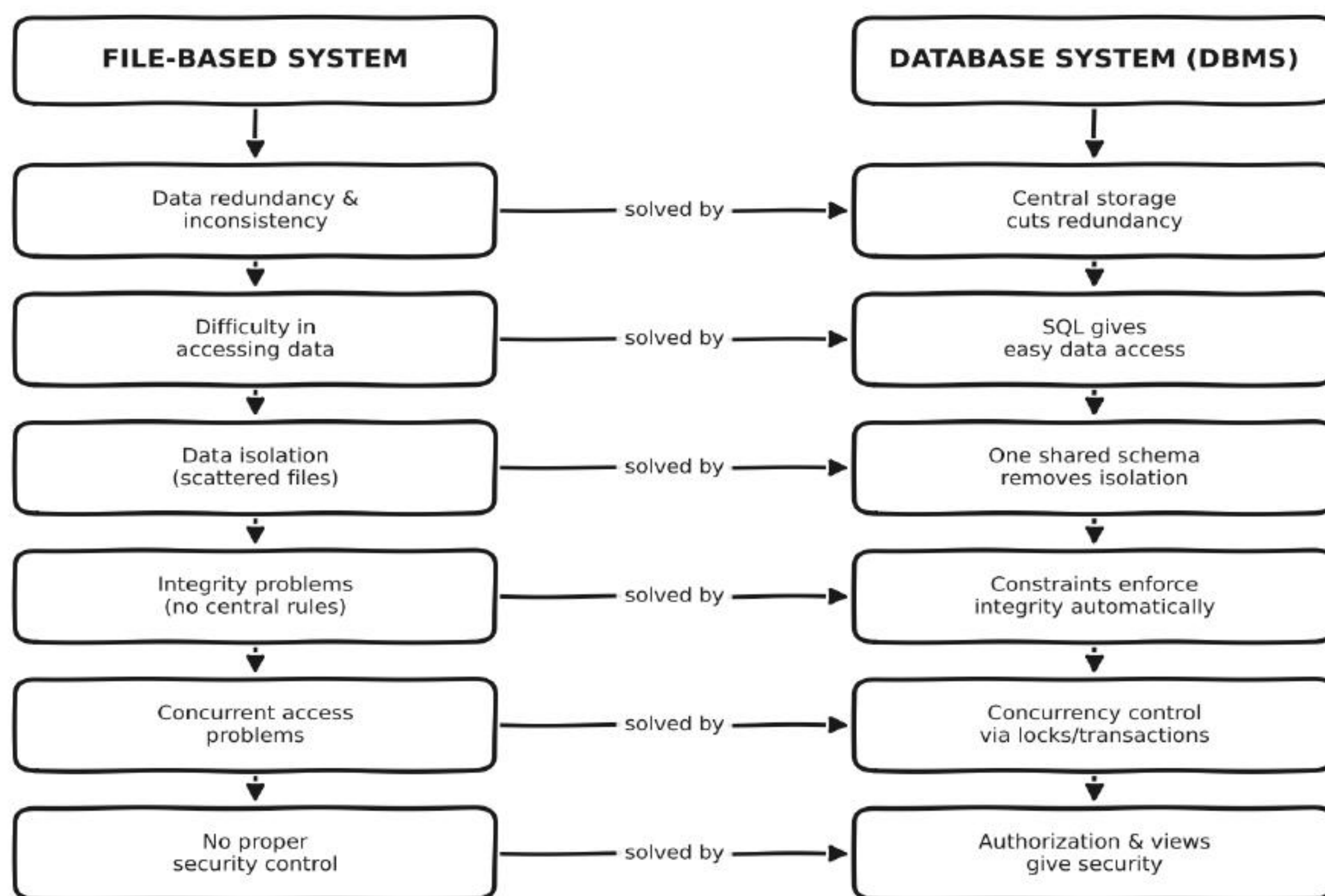


Q2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

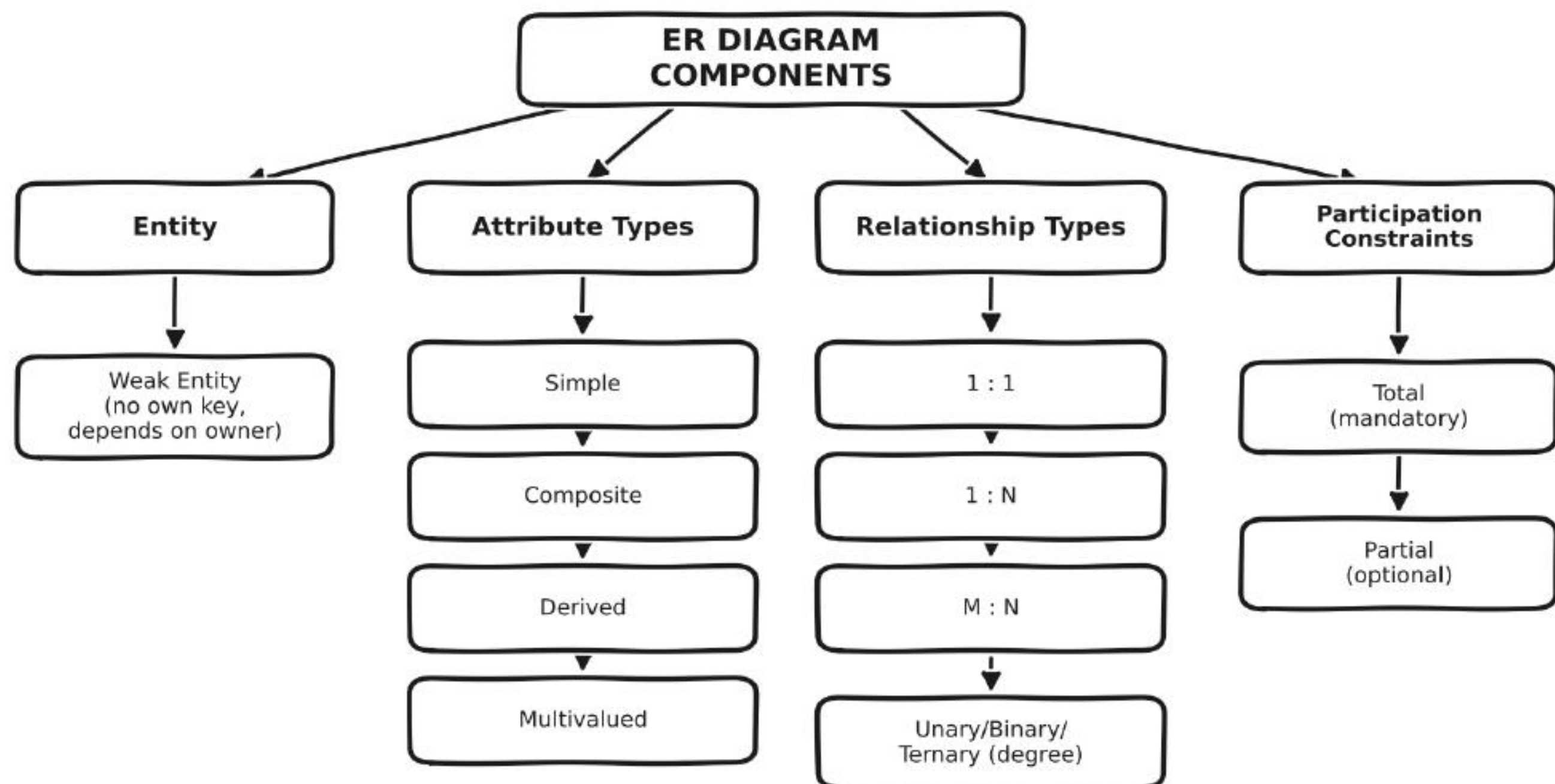




Q3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.

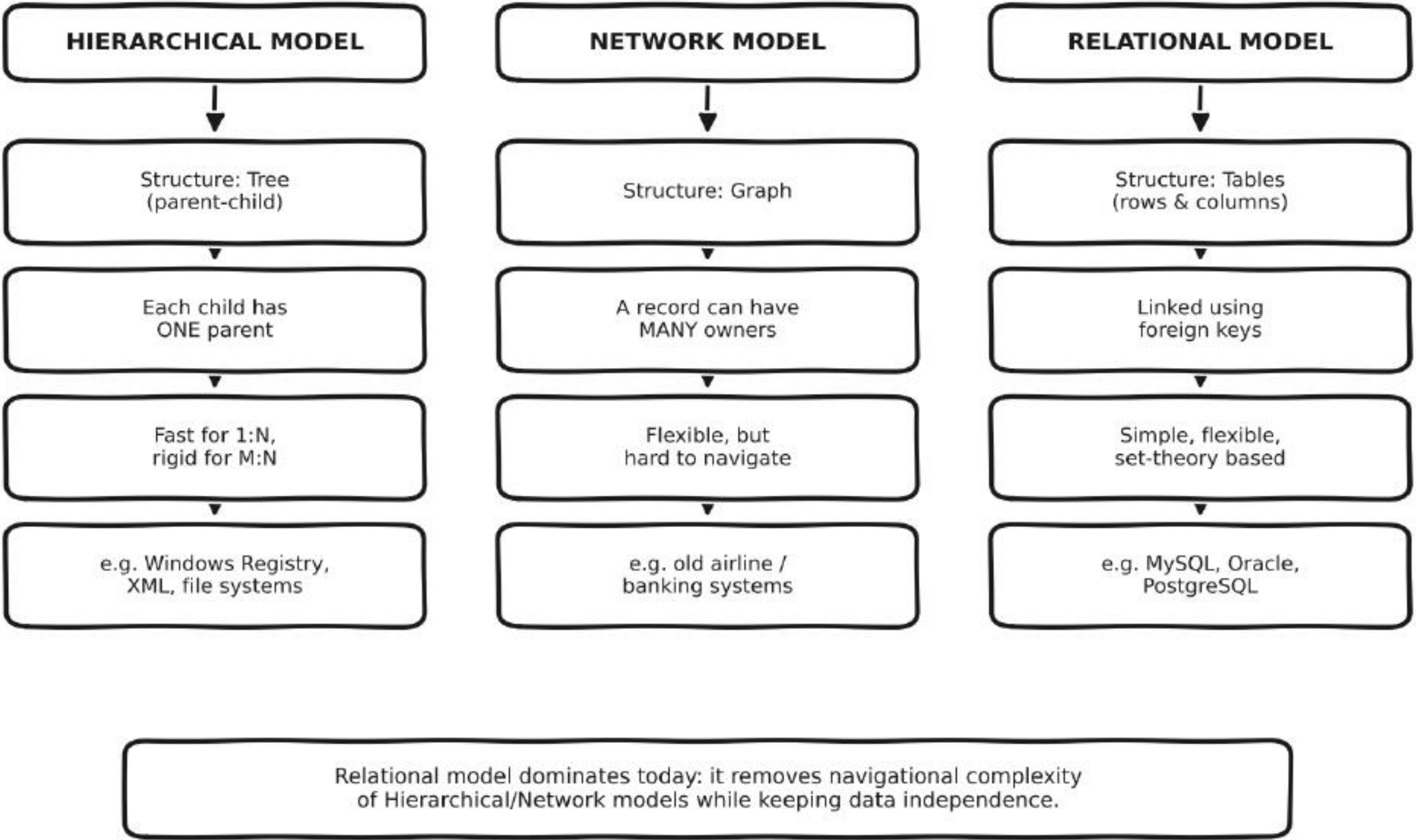


Q4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

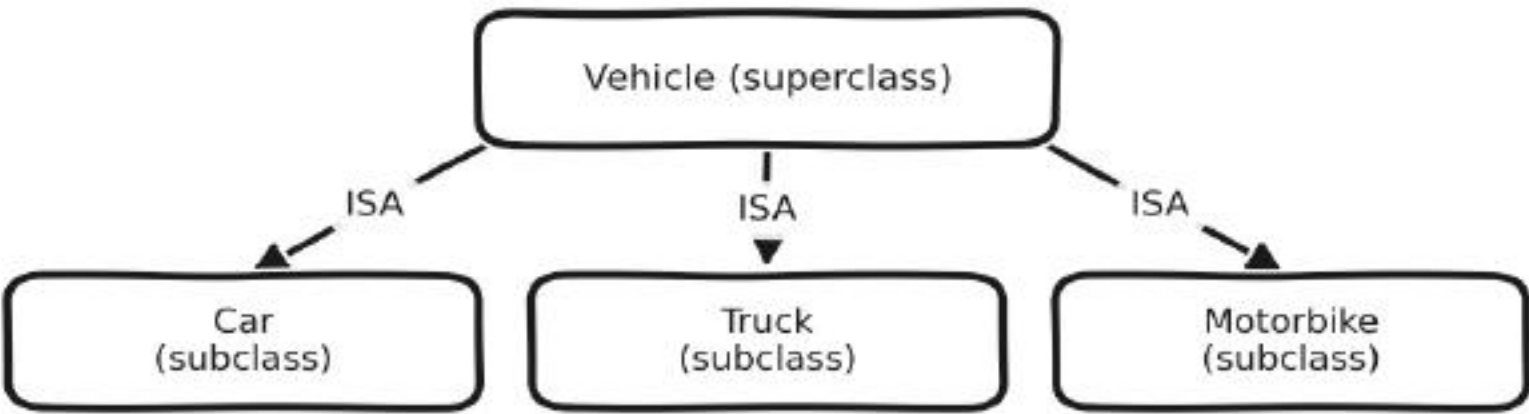
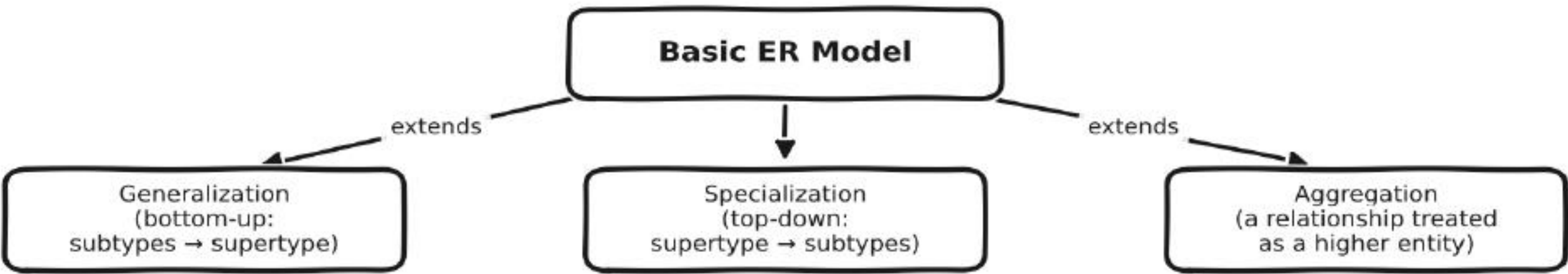




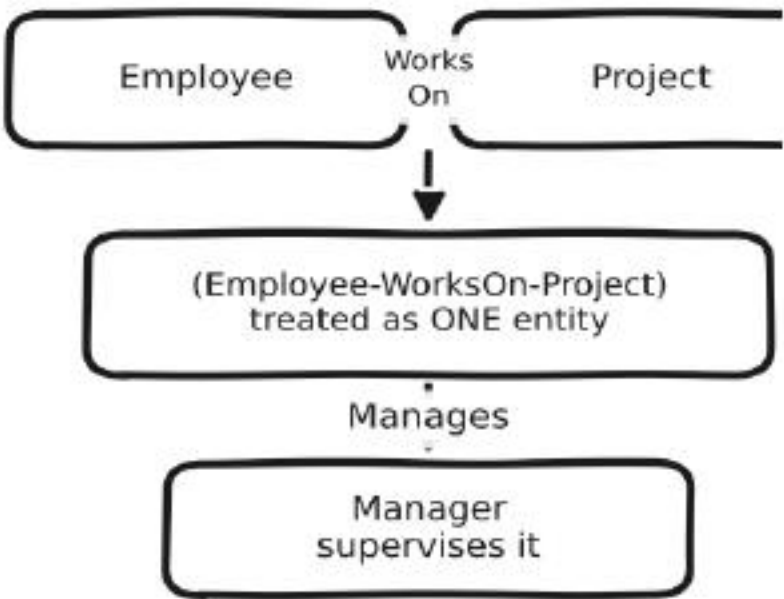
Q5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.



Q6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.



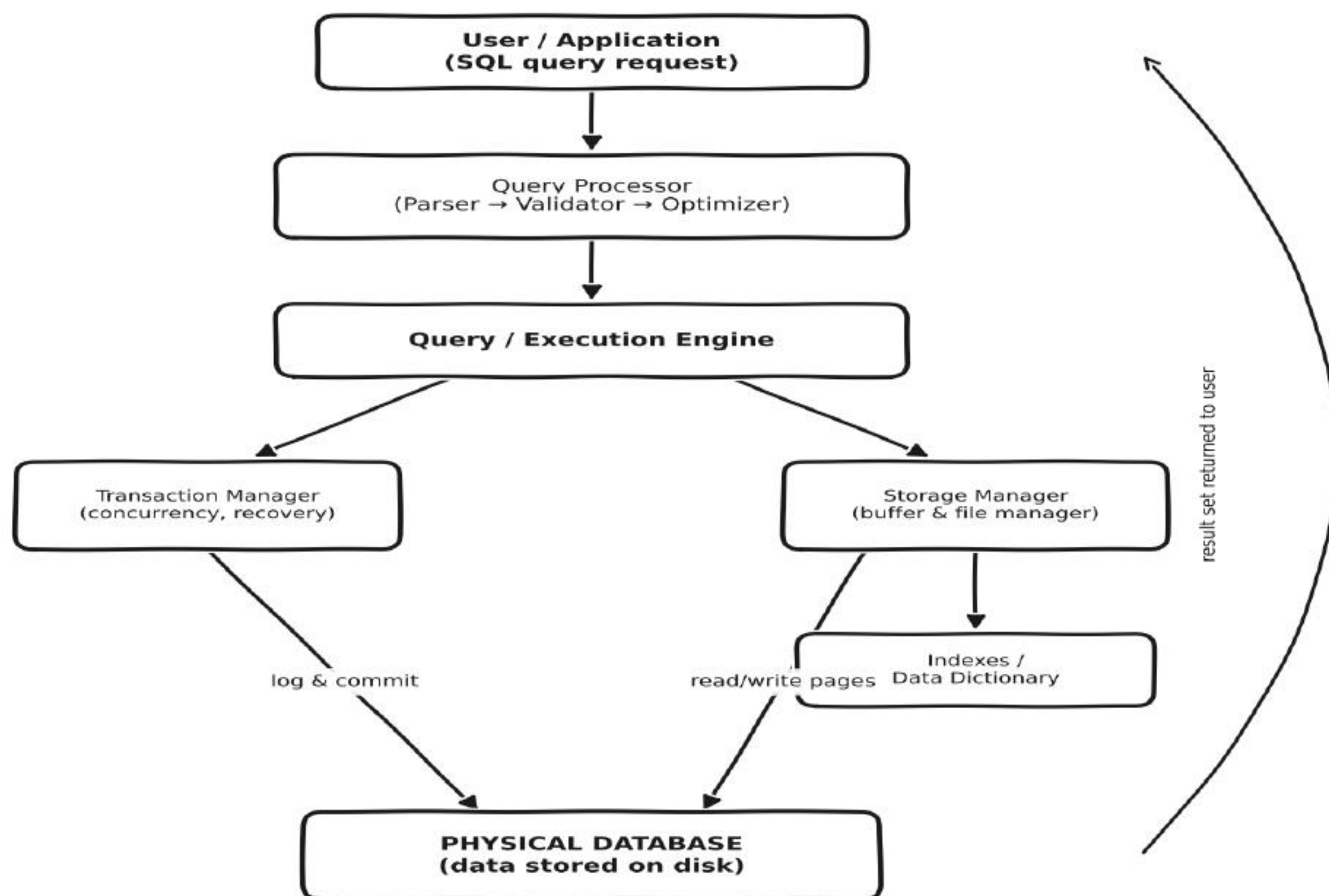
Generalization / Specialization example



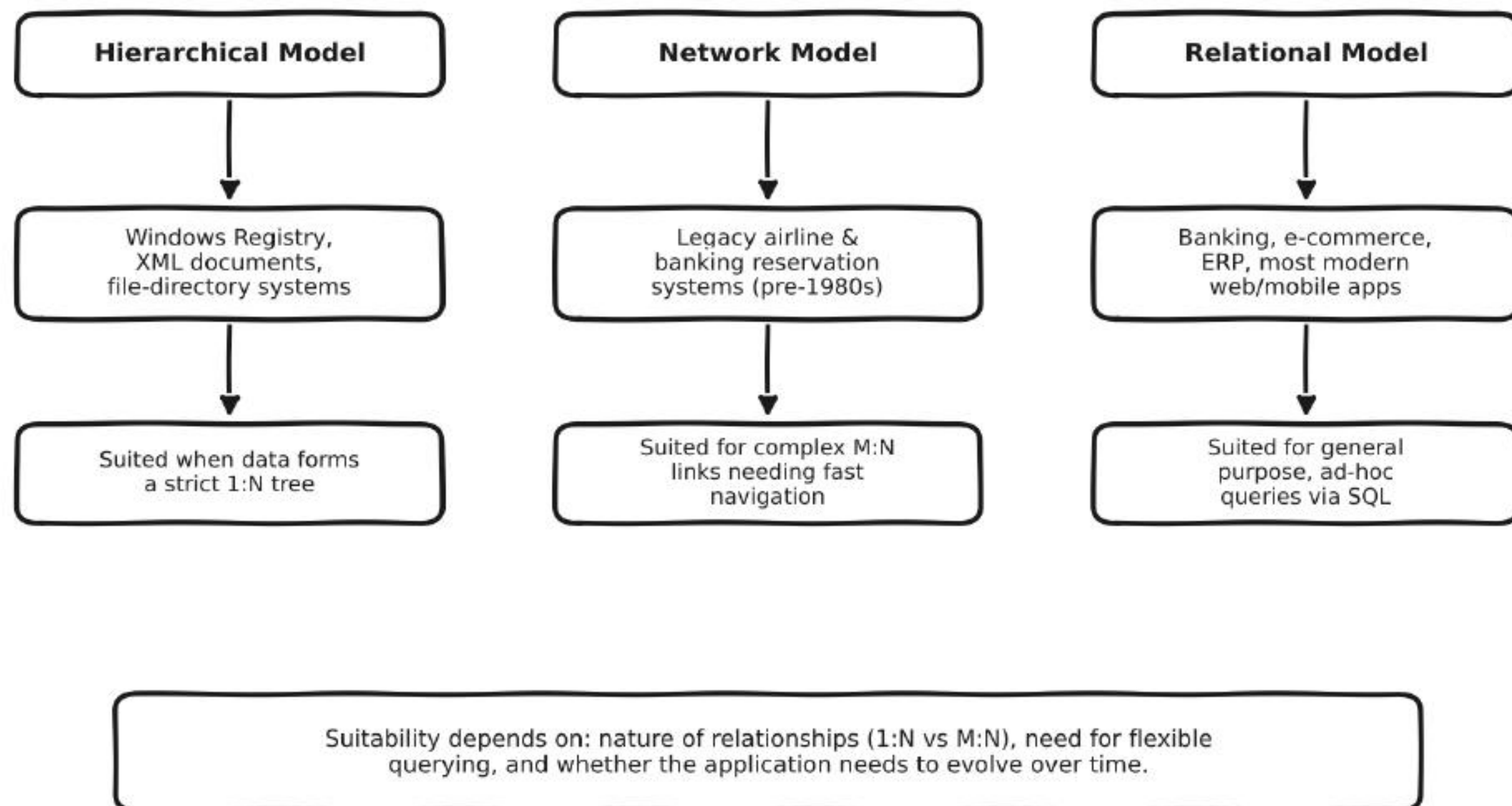
Aggregation example



Q7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.

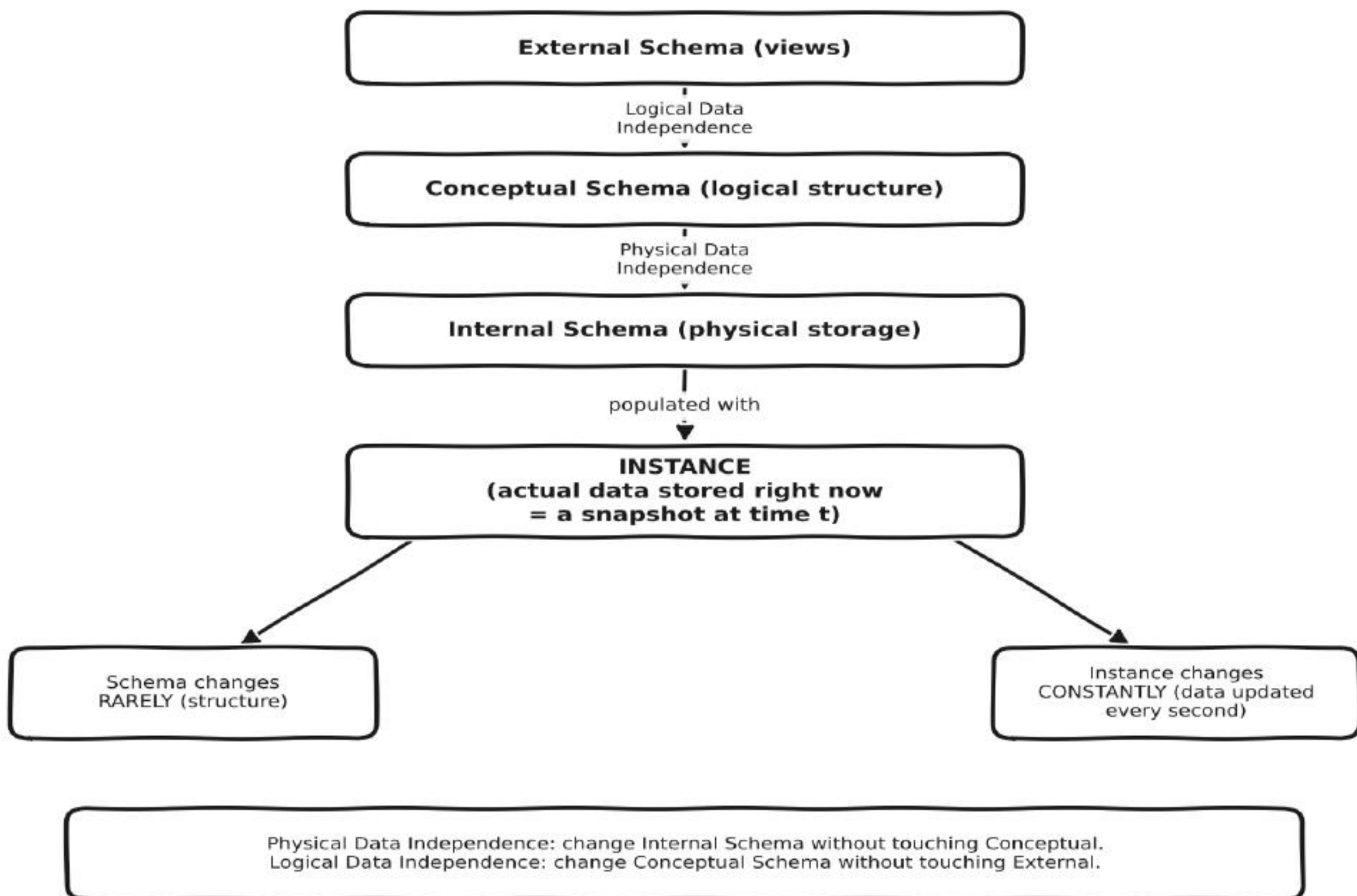


Q8. Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.





Q9. Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.



Q10. Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.

